

1 Data and Institutional Setting

Denmark offers an ideal setting for examining the labor market impacts of generative AI.

First, Danish workers have been at the forefront of generative AI adoption, with take-up rates comparable to those in the United States (Bick, Blandin and Deming, 2025; Humlum and Vestergaard, 2025; RISJ, 2024).

Second, Denmark’s labor market is highly flexible, with low hiring and firing costs—similar to those of the U.S.—which allow firms and workers to adjust employment in response to technological change (Botero et al., 2004; Dahl, Le Maire and Munch, 2013). Appendix C.1 details the wage-setting systems in our study occupations, showing that most workers engage in annual negotiations with their employers, providing regular opportunities to adjust earnings and hours in response to AI chatbot adoption.

Third, Denmark has exceptional infrastructure for tracking the adoption of new technologies. In particular, every Dane has a digital mailbox that Statistics Denmark can use to distribute survey invitations. We use this infrastructure to conduct two large-scale, representative surveys on AI chatbot adoption and work reorganization.

Finally, our partnership with Statistics Denmark allows us to link these surveys to administrative matched employer-employee data, providing a unique opportunity to analyze labor market effects such as changes in earnings, working hours, and job mobility.

Taken together, these factors make Denmark a prime setting for observing early labor market effects of generative AI, with insights that may extend to other advanced economies—including the U.S.—and an unparalleled data infrastructure linking workplace adjustments to administrative labor market outcomes.

1.1 Survey Data

This paper builds on two large-scale surveys on AI chatbot adoption, conducted in November–December of 2023 and 2024. The first survey provided the dataset for Humlum and Vestergaard (2025), which documented adoption patterns of ChatGPT, the dominant AI chatbot. This paper extends that dataset in two key ways. First, we link survey

responses to administrative labor market data after the introduction of ChatGPT, enabling us to assess impacts on earnings, hours worked, and job mobility. Second, we introduce a second survey round in 2024 that (i) broadens the scope to include all AI chatbots, including custom versions, (ii) provides extensive data on employer-led adoption initiatives, and (iii) documents changes in work organization alongside workers’ actual usage and perceived benefits of these tools. Appendix I outlines the 2024 survey, which is the primary focus of this paper.

1.1.1 Occupations

Our surveys focus on 11 occupations that are highly exposed to AI chatbots: *accountants, customer support specialists, financial advisors, HR professionals, IT support specialists, journalists, legal professionals, marketing professionals, office clerks, software developers, and teachers*. These occupations were selected based on three criteria: (i) they have at least one O*NET job task where AI chatbots can save time, as measured by the “Direct Exposure (E1)” metric from Eloundou et al. (2024); (ii) they are captured by a well-defined set of ISCO codes; and (iii) they contain a sufficient number of workers for statistical analysis. Humlum and Vestergaard (2025) details the selection and empirical measurement of these occupations. Together, these 11 occupations comprise the segments of the labor market where the impacts of AI chatbots are likely to emerge first and most strongly. Our analysis focuses on this set.

1.2 Register Data

We use several administrative registers at Statistics Denmark. Our matched employer-employee data come from the *Employment Statistics of Employees* (BFL), which records earnings, hours, occupation, and industry for all job spells in Denmark on a monthly basis from 2008 onward. This register is compiled by the Danish tax authorities and subsequently harmonized by Statistics Denmark. We complement this with demographics data on individuals from the *Population Register* (BEF), and wealth information from the *Personal Wealth Register* (FORMPERS). Finally, we draw firm financial data (e.g.,

annual revenue and employment) from the *Firm Statistics* (FIRM) Register. Because our survey was sent to workers identified in the register data by their (deidentified) social security numbers (*pnr*), all respondents can be matched to the register data. While labor market data in the BFL register are updated continuously, we only have firm financial data up to 2022. For that reason, we focus our effect analyses on labor market outcomes.

1.3 Survey Sample

We invited 115,000 workers to participate in each of our survey rounds in 2023 and 2024. The registers at Statistics Denmark enabled us to target these invitations by occupation and workplace. In particular, for each of our 11 occupations, we conducted a workplace-based sampling procedure, first drawing a random set of workplaces within each occupation and then sampling all relevant workers at these workplaces. This sampling procedure maximizes the statistical power of the workplace-level analyses while keeping our sample representative. Appendix B.1 details the sampling protocol. We sent three reminders per survey round, two by e-mail and one by text. The invitation letters are in Appendix H. We received about 25,000 valid and complete responses to each survey. Appendix B.2 details our survey response rates. While our main analysis focuses on responses from the 2024 round, we use the 2023 round to examine early adopters.

1.3.1 Representativeness and Response Quality

Appendix B.2.1 conducts several checks on the representativeness and quality of our survey responses. These analyses extend the checks in Humlum and Vestergaard (2025) to the 2024 survey round. First, we ensure that our sample represents the population based on observables, including age, gender, experience, earnings, and wealth. Second, following Dutz et al. (2025), we use randomized participation incentives to show that our findings are also balanced on workers' latent willingness to participate in the survey. Finally, we cross-check that the survey responses align with variables that are also recorded in the administrative registers.

2 The Currents: Adoption and Work Reorganization

In this section, we study how AI chatbots are integrated into the workplace. Section 2.1 describes employers’ chatbot initiatives, showing that even as chatbots have become widespread, otherwise similar firms often adopt markedly different approaches. In Section 2.2, we examine how these strategies reflect in workers’ chatbot use and its reported benefits. Finally, in Section 2.3, we study how AI chatbots reshape the organization of work, including the creation of new job tasks. Together, these results document the rapid currents of adoption and work reorganization set in motion by the arrival of AI chatbots.

2.1 Employer Initiatives

Figure 1, Panel (a), shows the prevalence of employer usage policies for AI chatbots across our 11 occupations. Firms now broadly embrace AI chatbots: about 43% explicitly encourage their use, another 21% allow it, while only about 6% explicitly prohibit it.⁴ This widespread adoption of AI chatbots in the workplace is consistent with their general-purpose features (Eloundou et al., 2024).

Employers not only set usage policies but also invest directly in chatbot adoption. Panel (b) shows the chatbot packages adopted by different employers. Among firms that encourage chatbot use, 61% have an enterprise chatbot solution, 39% provide training, and 29% offer both. Notably, enterprise chatbots and training are present even in some workplaces that do not encourage use.

Which firms have adopted different AI chatbot initiatives? Table E.1 shows that firms encouraging AI chatbot use tend to be slightly younger (each additional 10 years of age is associated with a 1.4 percentage point decrease in encouragement), more productive (a doubling of productivity corresponds to a 4.7 percentage point increase), and more likely to be privately owned (associated with a 4.1 percentage point increase). Yotzov et al.

⁴The widespread embrace of AI chatbots marks a shift from early responses to ChatGPT, when many employers restricted its use due to concerns about data confidentiality and output accuracy (Humlum and Vestergaard, 2025). Although outright bans are now rare, they persist in some occupations involving sensitive data (e.g., financial advisors) or requiring high factual accuracy (e.g., legal professionals).